

DATA-260 Homework 1 — Poushali

Configuration

- SID4: 0571
- PORT_BASE: 8571
- PREFIX: s0571
- SEED: 571
- VERIFY_SEED: 260571
- DOMAIN_ID: 3
- Assigned domain: Grocery supply and recall notices
- Hardware: Intel Core i5-1135G7, 15.8 GB RAM, Intel Iris Xe Graphics, Windows 11 Home Single Language
- Python: 3.12.10
- Local model: qwen3:8b through Ollama
- Tagged commit hash: fill in after the final hw1 tag is created

Part 1 — HTML, JavaScript, and deployment

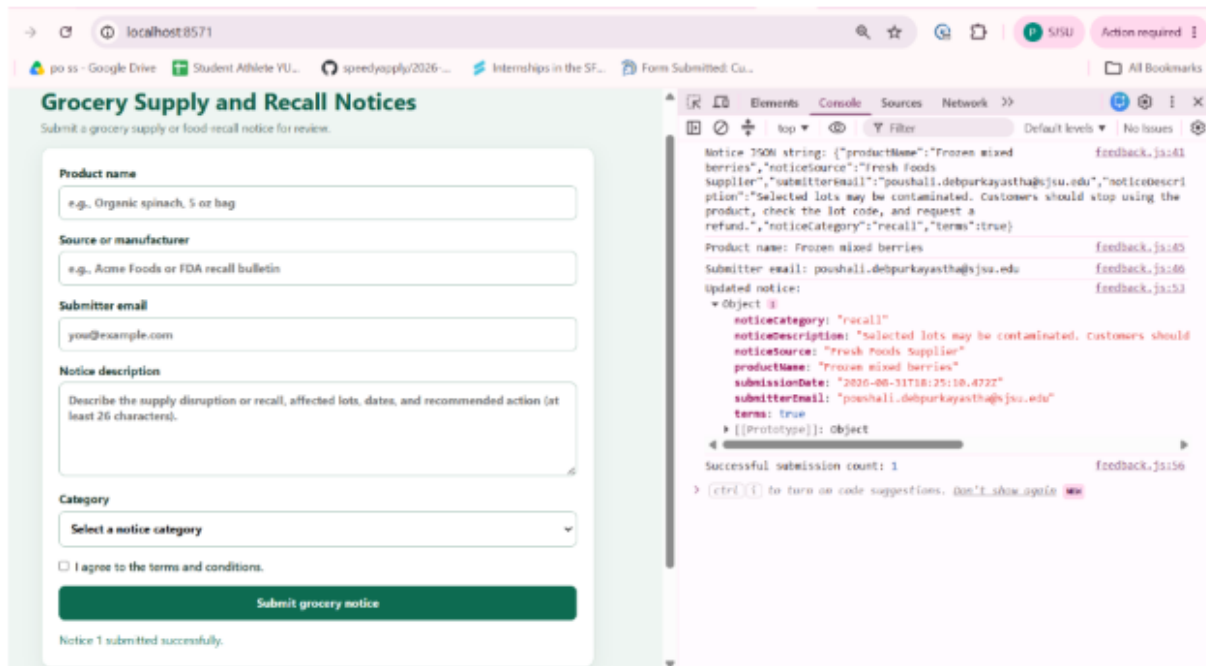
DOMAIN_SCHEMA.md defines the GroceryNotice entity before the page implementation. index.html is titled HW1-Poushali and uses the largest heading for the assigned domain. Its form has required product, source, submitter email, description, category, and terms controls. The product field has a descriptive placeholder and autofocus. The category has four domain values and the button says “Submit grocery notice.”

feedback.js uses an arrow function to reject descriptions of 25 or fewer characters and to reject an unchecked terms box. Successful data is stringified and parsed as JSON; destructuring extracts productName and submitterEmail; the spread operator adds an ISO submissionDate; and a closure counts successful submissions.

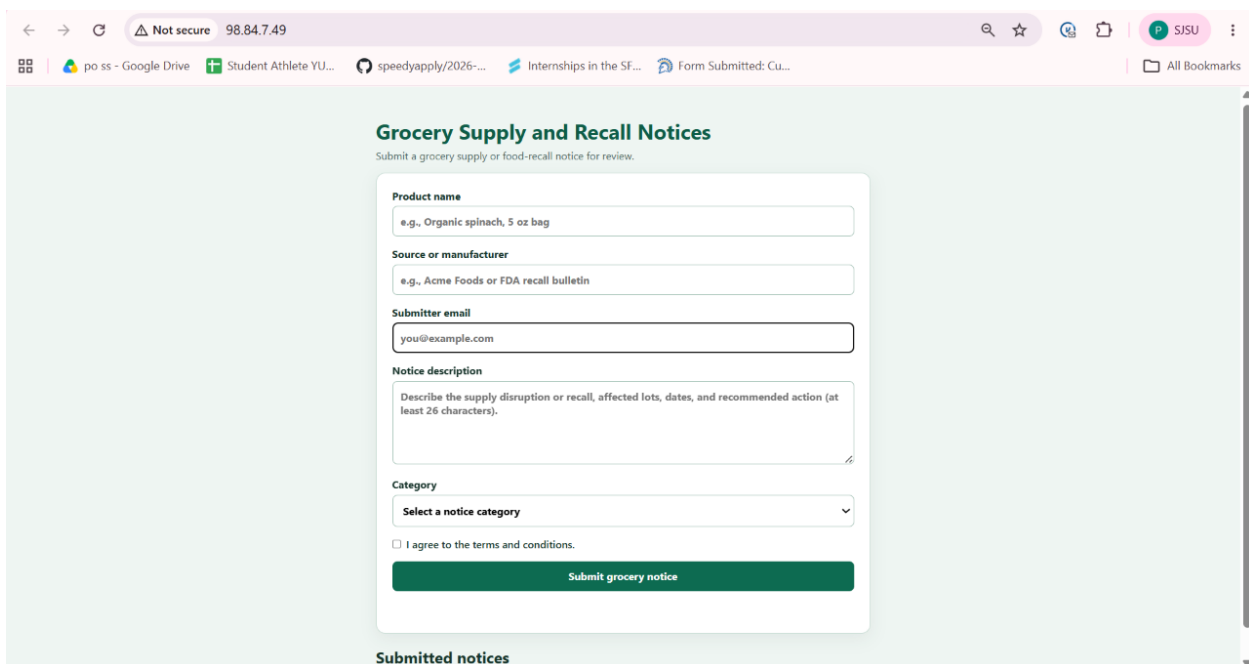
Local Docker command:

```
docker build -t data260-hw1-0571 .
docker run --rm -d --name data260-hw1-0571 -p 8571:80 data260-hw1-0571
```

Local screenshot (screenshots/local-console.png): the form at <http://localhost:8571> with the browser console after a valid submit.



The ECS deployment is defined by `ecs-task-definition.json` and `deploy_ecs.ps1`. It uses one Fargate task, port 80, public IP enabled, and an HTTP security-group rule. Public IP: `http://98.84.7.49`



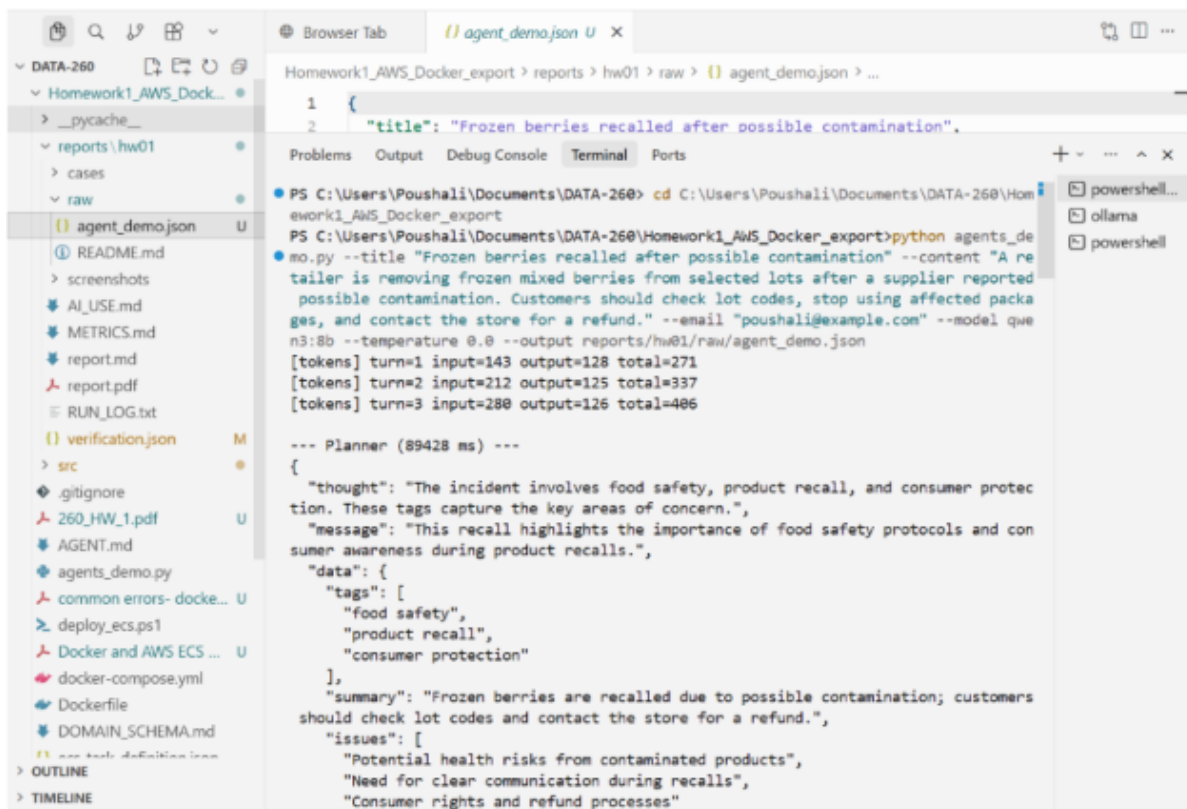
Part 2 — Agentic AI

agents_demo.py sends the supplied title and content through Planner, Reviewer, and Finalizer agents. The adapter requests JSON and the coercion layer enforces exactly three tags and a summary of at most 25 words. No grocery keywords are hardcoded into tag selection; candidates are derived from the input text.

Exact command:

```
python agents_demo.py --title "Frozen berries recalled after possible contamination" --content "A retail
```

Planner, Reviewer, Finalizer, and Publish JSON screenshots:



```
Homework1_AWS_Docker_export > reports > hw01 > raw > {} agent_demo.json > ...
1 {
2   "title": "Frozen berries recalled after possible contamination",
3 }

Problems Output Debug Console Terminal Ports
PS C:\Users\Poushali\Documents\DATA-260> cd C:\Users\Poushali\Documents\DATA-260\Homework1_AWS_Docker_export
PS C:\Users\Poushali\Documents\DATA-260\Homework1_AWS_Docker_export>python agents_demo.py --title "Frozen berries recalled after possible contamination" --content "A retailer is removing frozen mixed berries from selected lots after a supplier reported possible contamination. Customers should check lot codes, stop using affected packages, and contact the store for a refund." --email "poushali@example.com" --model qwen3:8b --temperature 0.0 --output reports/hw01/raw/agent_demo.json
[tokens] turn=1 input=143 output=128 total=271
[tokens] turn=2 input=212 output=125 total=337
[tokens] turn=3 input=280 output=126 total=406

--- Planner (89428 ms) ---
{
  "thought": "The incident involves food safety, product recall, and consumer protection. These tags capture the key areas of concern.",
  "message": "This recall highlights the importance of food safety protocols and consumer awareness during product recalls.",
  "data": {
    "tags": [
      "food safety",
      "product recall",
      "consumer protection"
    ],
    "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
    "issues": [
      "Potential health risks from contaminated products",
      "Need for clear communication during recalls",
      "Consumer rights and refund processes"
    ]
  }
}
```

Browser Tab: `/agent_demo.json`

Homework1_AWS_Docker_export > reports > hw01 > raw > `/agent_demo.json` > ...

```
1 {
2   "title": "Frozen berries recalled after possible contamination",
3 }
```

Problems Output Debug Console Terminal Ports

--- Planner (89428 ms) ---

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{
  "thought": "The incident involves food safety, product recall, and consumer protection. These tags capture the key areas of concern.",
  "message": "This recall highlights the importance of food safety protocols and consumer awareness during product recalls.",
  "data": {
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      "consumer protection"
    ],
    "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
    "issues": [
      "Potential health risks from contaminated products",
      "Need for clear communication during recalls",
      "Consumer rights and refund processes"
    ]
  }
}
```

--- Reviewer (38321 ms) ---

```
{
  "thought": "The content aligns with the provided tags and summary. Issues are relevant and specific. No major repairs needed.",
  "message": "Content is appropriate and well-structured with relevant tags and summary.",
  "data": {
    "tags": [
      "food safety",
      "product recall",
      "consumer protection"
    ],
    "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
    "issues": [
      "Potential health risks from contaminated products",
      "Need for clear communication during recalls",
      "Consumer rights and refund processes"
    ]
  }
}
```

powerShell... ollama powerShell

Browser Tab: `/agent_demo.json`

Homework1_AWS_Docker_export > reports > hw01 > raw > `/agent_demo.json` > ...

```
1 {
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```

Problems Output Debug Console Terminal Ports

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    "tags": [
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      "product recall",
      "consumer protection"
    ],
    "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
    "issues": [
      "Potential health risks from contaminated products",
      "Need for clear communication during recalls",
      "Consumer rights and refund processes"
    ]
  }
}
```

--- Finalized Output ---

```
{
  "thought": "The transcript provides consistent tags and summary, with clear issues identified. No changes are needed.",
  "message": "The recall of frozen berries due to contamination is well-documented with appropriate tags and summary.",
  "data": {
    "tags": [
      "food safety",
      "product recall",
      "consumer protection"
    ],
    "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
    "issues": [
      "Potential health risks from contaminated products",
      "Need for clear communication during recalls",
      "Consumer rights and refund processes"
    ]
  }
}
```

powerShell... ollama powerShell

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Homework1_AWS_Dock...

__pycache__

reports\hw01

cases

raw

agent_demo.json

README.md

screenshots

AI_USE.md

METRICS.md

report.md

report.pdf

RUN_LOG.txt

verification.json

src

.gitignore

260_HW_1.pdf

AGENT.md

agents_demo.py

common errors- docke...

deploy_ecs.ps1

Docker and AWS ECS ...

docker-compose.yml

Dockerfile

DOMAIN_SCHEMA.md

see tools definition icons

OUTLINE

TIMELINE

Browser Tabagent_demo.json U X

Homework1_AWS_Docker_export > reports > hw01 > raw > {} agent_demo.json > ...

1 {
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Problems Output Debug Console Terminal Ports

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"message": "The recall of frozen berries due to contamination is well-documented with appropriate tags and summary.",
"data": {
"tags": [
"food safety",
"product recall",
"consumer protection"
],
"summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
"issues": [
"Potential health risks from contaminated products",
"Need for clear communication during recalls",
"Consumer rights and refund processes"
]
}
}
--- Publish Output JSON ---
{
"title": "Frozen berries recalled after possible contamination",
"email": "poushali@example.com",
"content": "A retailer is removing frozen mixed berries from selected lots after a supplier reported possible contamination. Customers should check lot codes, stop using affected packages, and contact the store for a refund.",
"agents": {
powershell...
ollama
powershell

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Homework1_AWS_Dock...

__pycache__

reports\hw01

cases

raw

agent_demo.json

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Homework1_AWS_Docker_export > reports > hw01 > raw > {} agent_demo.json > ...

1 {
2 "title": "Frozen berries recalled after possible contamination",
Problems Output Debug Console Terminal Ports

--- Publish Output JSON ---
{
"title": "Frozen berries recalled after possible contamination",
"email": "poushali@example.com",
"content": "A retailer is removing frozen mixed berries from selected lots after a supplier reported possible contamination. Customers should check lot codes, stop using affected packages, and contact the store for a refund.",
"agents": {
"transcript": [
{
"role": "Planner",
"content": "{\n\"tags\": [\n\"food safety\", \n\"product recall\", \n\"consumer protection\"], \n\"summary\": \"Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.\", \n\"issues\": [\n\"Potential health risks from contaminated products\", \n\"Need for clear communication during recalls\", \n\"Consumer rights and refund processes\"]}"
},
{
"role": "Reviewer",
"content": "{\n\"tags\": [\n\"food safety\", \n\"product recall\", \n\"consumer protection\"], \n\"summary\": \"Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.\", \n\"issues\": [\n\"Potential health risks from contaminated products\", \n\"Need for clear communication during recalls\", \n\"Consumer rights and refund processes\"]}"
},
{
"role": "Finalizer",
"content": "{\n\"tags\": [\n\"food safety\", \n\"product recall\", \n\"consumer protection\"], \n\"summary\": \"Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.\". \n\"issues\":
powershell...
ollama
powershell


```
1 {
2   "title": "Frozen berries recalled after possible contamination",
3   "finalizer": {
4     "thought": "The transcript provides consistent tags and summary, with clear issues identified. No changes are needed.",
5     "message": "The recall of frozen berries due to contamination is well-documented with appropriate tags and summary.",
6     "data": {
7       "tags": [
8         "food safety",
9         "product recall",
10        "consumer protection"
11      ],
12       "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
13       "issues": [
14         "Need for clear communication during recalls",
15         "Consumer rights and refund processes"
16      ]
17     },
18     "final": {
19       "tags": [
20         "food safety",
21         "product recall",
22         "consumer protection"
23       ],
24       "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
25       "issues": [
26         "Potential health risks from contaminated products"
27       ]
28     }
29   }
30 }
```

```
1 {
2   "title": "Frozen berries recalled after possible contamination",
3   "finalizer": {
4     "data": {
5       "tags": [
6         "Need for clear communication during recalls",
7         "Consumer rights and refund processes"
8       ],
9       "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
10      "issues": [
11        "Potential health risks from contaminated products",
12        "Need for clear communication during recalls",
13        "Consumer rights and refund processes"
14      ]
15     },
16     "final": {
17       "tags": [
18         "food safety",
19         "product recall",
20         "consumer protection"
21       ],
22       "summary": "Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.",
23       "issues": [
24         "Potential health risks from contaminated products",
25         "Need for clear communication during recalls",
26         "Consumer rights and refund processes"
27       ],
28       "latencies_ms": {
29         "Planner": 89428,
30         "Reviewer": 38321,
31         "Finalizer": 45103
32       },
33       "submissionDate": "2026-08-31T19:57:37Z"
34     }
35   }
36 }
```

PS C:\Users\Poushall\Documents\DATA-260\Homework1_AWS_Docker_export>

Q1 final tags: food safety, product recall, consumer protection

Q2 final summary: **Frozen berries are recalled due to possible contamination; customers should check lot codes and contact the store for a refund.**

Q3 Reviewer changed anything: **No. The Reviewer did not change the Planner's tags or summary.**

Planner proposes input-derived tags and a summary. Reviewer checks relevance, generic wording, and the word limit. Finalizer uses the transcript to publish the strict final object.

Part 3 — Non-determinism

The fixed case is saved in `cases/nondeterminism_input.json` and is unchanged for all runs. The required command is:

```
python run_nondeterminism.py --model qwen3:8b --runs-per-temperature 20
```

The command writes 20 temperature-0.7 and 20 temperature-0.0 results to `raw/nondeterminism_runs.json`, including tags and latency.

| Metric | Temp 0.7 | Temp 0.0 |

|---|---|---|

| Distinct tag sets | 6 | 1 |

| Tags in all 20 runs | none | food safety, product recall, consumer protection |

| Tags in exactly 1 run | consumer advisory, contamination recall, customer recall action, frozen berries recall | none |

| Latency p50 / p95 / p99 (ms) | 79930 / 97093 / 98471 | 85534 / 93497 / 93591 |

Two users with identical input may see different tags or summaries at temperature 0.7. This is acceptable for exploratory topic discovery, but not for recall, safety, compliance, or other decisions that must be reproducible.

Part 4 — Model client and token accounting

`src/model_client.py` is the sole Ollama call path and exposes `complete(messages, tools=None)`. Each response prints input, output, and total tokens; `exit` prints cumulative input/output tokens and turn count. `hw1_client.py` loads `AGENT.md`, which requires strict bullet-only review output.

Exact five-turn command:

```
python hw1_client.py --demo --model qwen3:8b
```



```

PS C:\Users\Poushali\Documents\DATA-260\python hw1_client.py --demo --model qwen3:8b
Enter a message, /stats for statistics, or /quit to exit.
you> Review the HTML form for required-field accessibility issues.
[tokens] turn=1 input=80 output=100 total=180
assistant> - Ensure all required fields have the 'required' attribute or are marked with an '<abbr title="Required">*</abbr>' or similar indicator.
- Verify that labels are properly associated with input fields using 'for' attribute or '<label>' wrapping.
elds.
- Make sure form fields have unique 'id' attributes for proper label association.
you> Review the JavaScript validation and identify one edge case.
[tokens] turn=2 input=198 output=23 total=221
assistant> - Edge case: If a user disables JavaScript, the form validation will not work, leaving required fields unchecked.
you> Suggest one small test for successful form submission.
[tokens] turn=3 input=240 output=24 total=264
assistant> - Test: Submit the form with all required fields filled in and verify the form data is sent to the server.
stats after turn 3:
{
  "turn_count": 3,
  "cumulative_input_tokens": 518,
  "cumulative_output_tokens": 147,
  "serialized_conversation_history_length": 1363
}
you> Explain whether the current submission counter is a closure.
[tokens] turn=4 input=284 output=33 total=317
assistant> - The current submission counter is a closure if it is implemented using a function that captures and retains access to its lexical scope, including vari
outer functions.
you> Give a final two-item priority list for improving this code.
[tokens] turn=5 input=338 output=31 total=369
[tokens] turn=5 input=338 output=31 total=369
assistant> - Implement proper form field labeling and required indicators for better accessibility.
- Add server-side validation to handle cases where client-side validation may be bypassed.
stats after turn 5:
{
  "turn_count": 5,
  "cumulative_input_tokens": 1140,
  "cumulative_output_tokens": 211,
  "serialized_conversation_history_length": 1975
}
[cumulative] input=1140 output=211 turns=5

```

Stats after turn 3:

```

turn_count: 3
cumulative_input_tokens: 518
cumulative_output_tokens: 147
serialized_conversation_history_length: 1363

```

Stats after turn 5:

```

turn_count: 5
cumulative_input_tokens: 1140
cumulative_output_tokens: 211
serialized_conversation_history_length: 1975

```

Exit totals: input=1140 output=211 turns=5. Responses followed AGENT.md bullet-only review.

Prior conversation context is resent because Ollama chat completion is stateless between HTTP requests; the model needs earlier messages to answer follow-ups. A system prompt establishes behavior and constraints, while a user message supplies a request or data. Input tokens grow because the serialized history is included in each new request. The model context window and any application truncation or summarization eventually limit growth.

Verification and reproducibility

Run `python verify_hw01.py`; it writes `verification.json`. The report must be updated with real screenshots, agent output, metrics, `/stats` captures, and the final tagged commit hash before submission.